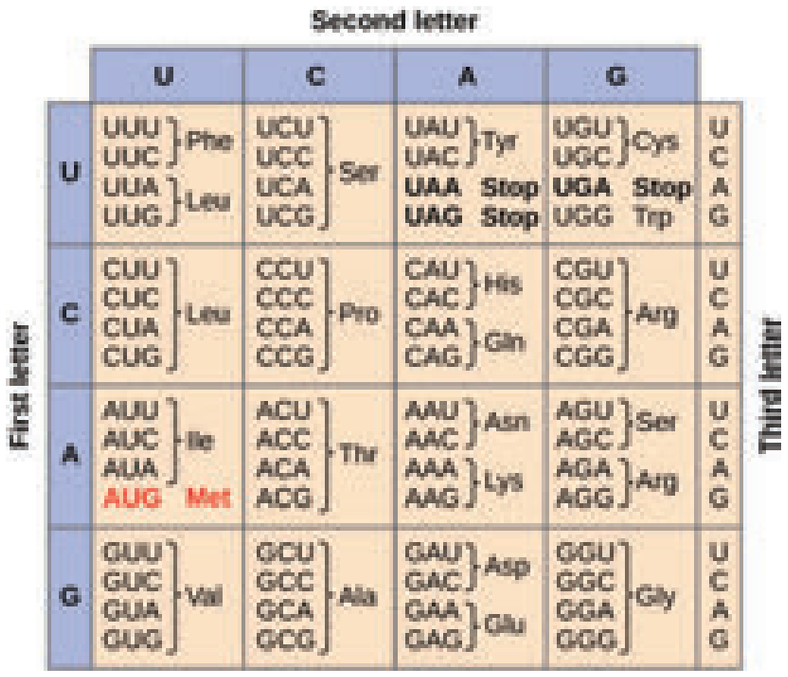


The 20 amino acids and their abbreviations	
Amino acid	3-letter abbreviation
Alanine	Ala
Arginine	Arg
Asparagine	Asn
Aspatic acid	Asp
Aspartic acid & Asparagine	Asx
Cysteine	Cys
Glutamine	Gln
Glutamic acid	Glu
Glutaine or Glutamic acid	Glx
Glycine	Gly
Histidine	His
Isoleucine	IIe
Leucine	Leu
Lysine	Lys
Methionine	Met

Table 12.1: Genetic code

HOTS

1. How many of the 64 codons can be made from the three nucleotides A, U, and C?
2. If codons were four bases long, how many codons would exist in a genetic code?

12.4 Translation



RNA is translated from the 5′ end of the molecule toward the 3′ end. Polypeptides are synthesized from the amino terminus toward the carboxyl terminus, by adding amino acids one by one to the carboxyl end. Not all the base sequences in an mRNA is translated into amino acid sequences of polypeptides. Initiation of polypeptide synthesis may begin hundreds of nucleotides from the 5′ – P terminus of the RNA. The section of untranslated RNA before the region encoding the first polypeptide chain is called a **leader**, which in some cases contains regulatory sequences that influence the rate of protein synthesis. The major events in translation are (Figure 12.4).

1. An mRNA binds to the surface of a protein synthesizing particle, the **Ribosome**.
2. The tRNA – amino acid complexes (made by the aminoacyl tRNA synthetases) bind sequentially, one by one, to the mRNA molecule that is attached to the ribosome.

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